

**A
REPORT
ON
INDUSTRIAL TRAINING ON
“PYTHON PROGRAMMING”**

Taken at



CODSOFT PVT LTD

In partial fulfillment of the requirement for the award of degree of

Bachelor of Technology

In

Information Technology



(Session 2025-26)

Duration: From 25/07/2025 to 25/08/2025

Submitted to:
Ms. Preeti Sharma

Submitted by:
RITIK AGARWAL
24EJCIT158
III Semester (Section C)

**Department of Information Technology
Jaipur Engineering College & Research Centre, Jaipur
Rajasthan Technical University
2025-26**



CERTIFICATE

OF COMPLETION
PROUDLY PRESENTED TO

RITIK AGARWAL

has successfully completed 4 weeks of a virtual internship program in

Python Programming

with wonderful remarks at **CODSOFT** from 25/07/2025 to 25/08/2025.

We were truly amazed by his/her showcased skills and invaluable contributions to the tasks and projects throughout the internship.



Founder



MSME
MICRO, SMALL & MEDIUM ENTERPRISES
सूक्ष्म, लघु एवं मध्यम उद्यम

CERTIFICATE

This is to certify that the report of the training submitted is the outcome of the practical training done at “**CodSoft**” in “**PYTHON PROGRAMMING**” is carried out by **Ritik Agarwal** bearing RTU Roll No.: **24EJCIT158** under the guidance and supervision of “**Mr. Manoj Dubey**” for the award of Degree of **Bachelor of Technology (B.Tech.) in Information Technology from Jaipur Engineering College & Research Centre, Jaipur (Raj.), India affiliated to Rajasthan Technical University, Kota** during the academic year 2024-2025.

To the best of the my knowledge the report

- i) Embodies the work of the candidate.
- ii) Has duly been completed.
- iii) Fulfills the requirement of the ordinance relating to the bachelor of technology degree of the Rajasthan technical University and
- iv) Is up to the desired standard for the purpose of which is submitted.

Dr. Smita Agarwal

Head of Department
Information Technology
JECRC, Jaipur

Mr. Piyush Gautam

Assistant Professor & TPO
Information Technology
JECRC, Jaipur

Ms. Preeti Sharma

Assistant Professor
Information Technology
JECRC, Jaipur

DECLARATION

I hereby declare that the report entitled “PYTHON PROGRAMMING” has been carried out and submitted by the undersigned to the Jaipur Engineering College & Research Centre, Jaipur (Rajasthan) in an original work, conducted under the guidance and supervision of Mr. Manoj Dubey of CodSoft Pvt. Ltd, Jaipur.

The empirical findings in this report are based on the data, which has been collected by me. I have not reproduced from any report of the University neither of this year nor of any previous year.

I understand that any such reproducing from an original work by another is liable to be punished in a way the University authorities’ deed fit.

Place: Jaipur

Date:

RITIK AGARWAL

24EJCIT158

PREFACE

Bachelor of Technology in Information Technology is the Rajasthan Technical University course (Approved by AICTE) having duration of 4 years. As a prerequisite of the syllabus every student on this course has to take professional training from any Industry related to the stream for 60 days in order to complete his/her studies successfully. And it is required to submit the project report on the completion of it.

The main objective of this training is to create awareness regarding the application of theories in the practical world of Information Technology and to give a practical exposure of the real world to the student.

I, therefore, submit this project report of this training, which was undertaken at CodSoft Pvt. Ltd, Jaipur. It was a good experience to have training in a Reputed Organization in the Bigdata Technology in which company stands at international.

I feel great pleasure to present this project report. The report was prepared on the basis of the study carried out regarding the existing File Tracking System.

ACKNOWLEDGEMENT

“Any serious and lasting achievement or success, one can never achieve without the help, guidance and co-operation of so many people involved in the work.

It is my pleasant duty to express my profound gratitude and extreme regards and thanks to **Mr. Arpit Agarwal and Dr. V.K. Chandna** who gave me an opportunity to take professional training in CodSoft Pvt. Ltd, Jaipur.

I am indebted towards Mr. Manoj Dubey who has allotted this project and his precious time and advice during the training period, which is imminent to the report.

I would like to express deep gratitude to **Dr Smita Agrawal**, Head of Department (Information Technology), Jaipur Engineering College & Research Centre, Jaipur (Rajasthan) without whose permission the training would not be possible. I would also like to thank **Mr. Piyush Gautam**, Training & Placement Officer and **Mr. Rohit Chhabra**, PTS Coordinator, Department of Information Technology JECRC, Jaipur who recommended me for this training.

Last but not the least, I am grateful to my beloved parents whose blessings and inspirations are always with me. I am heartily thankful to those people who are involved directly or indirectly in this project for encouraging me whenever I needed their help in spite of their busy schedule.

RITIK AGARWAL

24EJCIT158

Chapter Index

S. No.	TITLE	PAGE NO.
	Certificate	i
	Declaration	ii
	Preface	iii
	Acknowledgement	iv
	Chapter Index	v
	Figure Index	vii
	Company Profile	viii
1.	Introduction	1
	1.1 Purpose	1
	1.2 Scope	2
	1.3 Features	2
2.	What is Python Programming?	4
	2.1 Conceptual Relevance	5
3.	Tools Used	6
	3.1 Visual Studio Code	6
	3.2 Randon Module	7
	3.3 Integrated Development Environment (IDE)	7
	3.4 Command Prompt/ Terminal	9

	3.5 System Hardware and OS	10
	3.6 Text Formatting & Documentation Tools	11
4.	Project Description	12
	4.1 Project Title	12
	4.2 Objective	12
	4.3 Description of Project	13
	4.4 Tools and Technologies Used	13
	4.5 Modules of the Project	14
	4.6 Algorithm/ Working Steps	15
	4.7 Sample Output	16
	4.8 Learning Outcomes	16
5.	Screenshots	18
	Conclusion	20
	Future Scope	23
	References	26

Figure Index

S. No	Title	Page Number
5.1	Screenshots of Projects	18
5.2	Output Of Projects	19

COMPANY PROFILE

CODSOFT is a rapidly growing IT organization dedicated to fostering innovation and practical learning in the field of software development and emerging technologies. Established with the vision of bridging the gap between academic education and industry requirements, CODSOFT offers hands-on learning experiences, internships, and real-world project opportunities to students and professionals aspiring to build successful careers in the technology sector.

Since its inception, CODSOFT has aimed to create a dynamic learning environment where creativity, collaboration, and technical expertise come together. The organization focuses on developing future-ready professionals by providing them with live project exposure and mentorship from experienced industry experts. CODSOFT's primary objective is to empower individuals with the necessary skills and confidence to succeed in the ever-evolving IT landscape.

CODSOFT is known for its extensive internship programs in domains such as **Python Programming, Web Development, Artificial Intelligence, Data Science, Java, Machine Learning, and UI/UX Design**. These programs are carefully structured to provide participants with both theoretical understanding and practical application, helping them gain real-time experience in solving complex technological challenges.

The company's strength lies in its commitment to high-quality training, mentorship, and project-based learning. Every participant is encouraged to work on innovative ideas, develop logical problem-solving abilities, and enhance coding efficiency through structured modules and guided practice. CODSOFT ensures that learners not only understand technical concepts but also develop teamwork, communication, and analytical skills essential for professional success.

In addition to training and internships, CODSOFT also contributes to technological advancement by offering consultancy, software development, and automation solutions to clients worldwide. With a strong focus on excellence, ethics, and continuous improvement, CODSOFT has earned a reputation as a reliable platform for aspiring developers and technologists.

Key Features of CODSOFT:

- Offers **100% practical, project-oriented training and internships.**
- Provides expert guidance from **qualified mentors and industry professionals.**
- Focuses on **skill development, innovation, and real-world application.**
- Encourages **self-learning and professional growth** through modern teaching methodologies.
- Operates with a mission to **build future technologists** ready for global IT challenges.

In conclusion, CODSOFT stands as a centre of excellence for young learners and professionals seeking to gain hands-on experience in programming and software development. Through its innovative training programs and strong educational values, the organization continues to contribute to the development of skilled and confident IT professionals for the future.

CHAPTER 1

INTRODUCTION

1.1 PURPOSE

The purpose of this project, titled “**Rock–Paper–Scissors Game using Python,**” is to demonstrate the fundamental programming concepts and logical thinking skills that can be achieved through the Python programming language. Python is one of the most versatile and beginner-friendly languages widely used in various domains such as web development, automation, data analysis, and software development. Through this project, the objective is to apply the core principles of Python to design a simple, interactive, and user-friendly application that performs logical decision-making based on user input.

This project helps to understand how Python can be used to develop small yet effective console-based applications. The game involves two players — the user and the computer — where both select one option among Rock, Paper, or Scissors. The program then determines the winner using simple conditional logic. This process allows students to gain practical experience in using **control statements (if–else), input/output functions, loops, dictionaries,** and the **random module** of Python.

The project aims to fulfil the following objectives:

- To develop a basic understanding of how to take **user input** and process it programmatically.
- To learn the use of **conditional and logical operators** for making decisions in code.
- To understand how **Python modules** like random can be used to introduce dynamic and unpredictable behaviour.
- To gain hands-on experience in writing **error-free, structured, and modular code.**
- To improve problem-solving skills by implementing real-world logic into a computer program.

This project also highlights Python’s simplicity and readability, which make it ideal for learning and implementing basic algorithms. The concise syntax of Python helps in focusing on the logic of the problem rather than the complexity of the language, making it easier for beginners to grasp programming concepts effectively.

Moreover, creating such a game encourages logical reasoning and computational thinking. It demonstrates how programs can be designed to take inputs, compare conditions, and provide

appropriate outputs — which are the fundamental aspects of programming. The Rock–Paper–Scissors game serves as a foundation for developing more complex applications in the future by using similar programming logic and structures.

In conclusion, the purpose of this project is to utilize Python programming for building an interactive console-based game that enhances programming knowledge, logical thinking, and practical understanding of basic coding concepts. It represents how programming can be used not only for solving technical problems but also for creating engaging and educational applications.

1.2 SCOPE

The scope of this project, titled “Rock–Paper–Scissors Game using Python,” is to apply the fundamental concepts of Python programming to develop a simple, interactive console-based game. It focuses on implementing logical operations, conditional statements, and randomization to simulate a game between the user and the computer.

This project covers:

- Taking user input and displaying results using input/output functions.
- Applying conditional statements to determine the winner.
- Using the random module to generate computer choices.
- Implementing dictionaries for mapping user and system selections.
- Writing clean, structured, and error-free code for smooth execution.

The project helps strengthen problem-solving skills and provides a clear understanding of Python syntax, logic building, and program flow. Though limited to a simple text-based game, the same principles can be extended to develop more advanced applications and games in the future.

1.3 FEATURES

The “**Rock–Paper–Scissors Game using Python**” project incorporates a variety of features that showcase the fundamental aspects of Python programming and logical thinking. It serves as a simple yet practical example of how Python can be used to build interactive applications through concise and efficient coding techniques.

Key Features:

- **Interactive Gameplay:** The program allows the user to actively participate in a game against the computer by selecting between rock, paper, and scissors. This helps in understanding user input handling and interactive programming.
- **Randomized Computer Choice:** The use of Python's built-in random module enables the computer to make unpredictable choices, ensuring that each round of the game produces different results.
- **Decision-Making Logic:** The program uses conditional statements (if, elif, else) to compare user and computer choices and determine the winner, demonstrating how logic and comparison operators work in Python.
- **User-Friendly Output:** Clear and formatted output is displayed to show both the user's and computer's selections along with the final game result, providing an engaging and easy-to-understand interface.
- **Efficient Code Structure:** The project is written using clean and readable Python syntax, making it easy to modify, debug, and extend for future development.
- **Minimal Resource Requirement:** The program runs efficiently in any standard Python environment without requiring external libraries or complex setup, highlighting Python's simplicity and accessibility.
- **Educational Utility:** This project is ideal for beginners to practice programming logic, control flow, and modular design, serving as a stepping stone for developing larger applications or games in Python.

In summary, the project demonstrates how Python can be used to design a functional and interactive application through logical operations, randomization, and structured programming. It highlights the language's simplicity, flexibility, and suitability for both beginners and developers aiming to strengthen their programming foundation.

CHAPTER 2

WHAT IS PYTHON PROGRAMMING

Python is a high-level, interpreted, and general-purpose programming language that is widely used for developing a variety of applications. Created by Guido van Rossum in the late 1980s and officially released in 1991, Python has grown to become one of the most popular programming languages in the world due to its simplicity, readability, and versatility.

2.1 Conceptual Relevance

The **conceptual relevance** of the project titled “**Rock–Paper–Scissors Game using Python**” lies in its ability to combine theoretical programming concepts with real-time implementation, thereby enhancing the understanding of logical thinking and algorithmic problem-solving. This project is designed to demonstrate how basic principles of programming can be applied to create an interactive and functional computer program using simple logic and code structure.

The project is based on the widely known *Rock–Paper–Scissors* game, which follows a set of predefined rules and outcomes. By converting this real-world concept into a digital program using Python, learners can practically understand the working of **conditional statements**, **random number generation**, **input/output handling**, and **data mapping**. Through this, students not only learn to code but also comprehend the reasoning and logic that form the backbone of software development.

In this project, **Python’s random module** is used to generate the computer’s choice, showcasing how programs can produce dynamic and unpredictable results. The use of **conditional statements** (if, elif, and else) helps determine the outcome by comparing the user’s and computer’s inputs. This illustrates the practical use of decision-making logic in programming, which is essential in all real-world applications ranging from games to automation systems.

Furthermore, this project introduces the concept of **data representation and mapping** by using Python dictionaries to associate user inputs (like ‘r’, ‘p’, and ‘s’) with their respective meanings (“Rock,” “Paper,” “Scissors”). Such mapping techniques are widely used in software applications for efficient data handling and retrieval.

From a conceptual standpoint, the project emphasizes **modular programming**, where the program is broken into smaller, manageable parts — input handling, logic processing, and output display. This approach enhances code readability, reusability, and maintenance, which are key elements of good programming practice.

The relevance of this project also lies in the development of **computational thinking** — the ability to approach problems systematically, identify logical patterns, and implement solutions through code. It encourages a hands-on understanding of how algorithms work, how programs make decisions, and how outputs are generated based on defined conditions.

Additionally, this project highlights the **real-world application of Python** as a beginner-friendly yet powerful programming language. It shows how Python can be used not just for simple console-based tasks but as a foundation for developing larger and more complex systems in areas like web development, automation, and artificial intelligence.

In summary, the conceptual relevance of this project is to strengthen the understanding of **core programming principles** and **logical problem-solving skills** through a simple, interactive, and enjoyable application. It bridges the gap between theoretical knowledge and practical execution, enabling learners to grasp how coding logic translates into working software, thereby laying a solid foundation for future technological learning and innovation.

CHAPTER 3

TOOLS USED

3.1 Visual Studio Code

Visual Studio Code (VS Code) is the primary tool used for developing and executing this Python project. It is a free, open-source, and lightweight **source code editor** developed by **Microsoft**, designed to support modern software development workflows. VS Code provides an efficient and user-friendly environment for writing, testing, and debugging Python programs.

The key advantage of Visual Studio Code lies in its **versatility** and **extensive extension support**. It allows integration with the **Python extension by Microsoft**, which provides essential features like **syntax highlighting**, **auto-completion**, **intelligent code suggestions (IntelliSense)**, **error detection**, and **debugging tools**. These features significantly enhance productivity and help developers write clean, efficient, and error-free code.

VS Code supports multiple programming languages, including **Python**, **C**, **C++**, **Java**, **JavaScript**, and **HTML/CSS**, making it ideal for beginners and professionals alike. It also provides **Git integration**, allowing version control directly within the editor. This helps in managing code revisions, collaborating with others, and maintaining a professional workflow similar to real-world software development environments.

For this Python project, Visual Studio Code was used to:

- Write and edit the program code efficiently.
- Execute and test the Python script using the integrated terminal.
- Debug and troubleshoot logical or syntax errors.
- Utilize Python extensions for better readability and performance analysis.
- Maintain a clear and organized workspace for project files.

In addition, VS Code offers customization options such as themes, layout settings, and extensions that can be tailored according to the developer's preference. Its smooth performance and adaptability make it one of the most popular development tools among programmers worldwide.

Overall, **Visual Studio Code** served as an ideal platform for developing the *Rock–Paper–Scissors Game in Python*, providing all the essential tools and a professional environment to implement, test, and refine the project efficiently.

3.2 Random Module

The **Random module** in Python is one of the most commonly used built-in libraries that helps in generating random numbers and making random selections. It is widely used in programs involving **games, simulations, data sampling, encryption, and statistical modeling**.

In this project — “*Rock, Paper, Scissors Game*” — the **Random module** plays a key role in allowing the **computer to make unpredictable choices** between Rock, Paper, or Scissors. This ensures that the game outcome is fair and unbiased, giving the user a realistic gaming experience.

Key Features of the Random Module

1. Random Selection:

The function `random.choice()` is used to pick a random element from a given list or sequence. In this project, it is used to let the computer randomly choose between [1, 2, 3], representing Rock, Paper, and Scissors.

2. Random Number Generation:

The module also supports other methods like `random.randint()` to generate random integers and `random.random()` to produce floating-point numbers between 0 and 1. These features make it highly useful in a wide range of applications.

Advantages in the Project

- Adds unpredictability and fairness to the game.
- Makes the computer’s move dynamic rather than fixed.
- Simulates real-world randomness, enhancing the user experience.

The **Random module** is a powerful and simple-to-use tool that adds variability and excitement to programs. In this project, it served as the backbone of the game’s logic, making the *Rock–Paper–Scissors* experience more engaging and authentic.

3.3 Integrated Development Environment (IDE)

An **Integrated Development Environment (IDE)** is a software application that provides comprehensive facilities to computer programmers for software development. It brings together all the essential tools needed to **write, test, debug, and run programs** within a single, user-friendly interface.

In this project, the IDE used is **Visual Studio Code (VS Code)**, one of the most popular and versatile development environments for Python and many other programming languages.

Key Features of an IDE

1. Code Editor:

Provides syntax highlighting, code completion, and indentation features that make coding faster and reduce syntax errors.

2. Debugger:

Helps in identifying and correcting logical or runtime errors by allowing developers to step through the code line by line.

3. Terminal/Command Line Integration:

Enables direct execution of Python programs without switching to a separate terminal, improving workflow efficiency.

4. Extensions and Plugins:

IDEs like VS Code support a variety of extensions, including Python interpreters, linters, and formatters that enhance development experience.

5. User Interface:

Offers an organized environment where files, folders, and output windows are managed efficiently.

Use in This Project

In the “*Rock, Paper, Scissors*” game project, the IDE was used for:

- Writing the Python code with clear syntax highlighting and auto-completion.
- Running and testing the program seamlessly.
- Debugging and checking output efficiently.

Advantages

- Simplifies the coding and debugging process.
- Increases productivity by providing multiple development tools in one platform.
- Offers an intuitive interface suitable for both beginners and professionals.

An **IDE** plays a vital role in modern software development by combining all essential programming tools under one environment. In this project, **Visual Studio Code** served as an effective IDE, making the Python coding process smoother, faster, and more efficient.

3.4 Command Prompt/ Terminal

The **Command Prompt** (in Windows) or **Terminal** (in Linux and macOS) is a powerful text-based interface used to interact with the operating system. It allows users to execute commands directly without the need for a graphical interface, making it an essential tool for developers and programmers.

In Python programming, the **Command Prompt/Terminal** is often used to **run Python scripts, install packages, and manage environments**, providing a simple and efficient way to work with the language.

Key Features

1. **Execution of Python Programs:**

Python files (.py) can be executed directly using the command `python filename.py`, allowing developers to quickly test and run their code.

2. **Package Management:**

The terminal supports commands like `pip install package_name` to install, update, or remove Python libraries and dependencies.

3. **Navigation and File Management:**

Users can navigate between folders, create or delete directories, and manage files efficiently using simple commands.

4. **Integration with IDEs:**

The terminal is often integrated within IDEs such as **Visual Studio Code**, enabling seamless execution of code without leaving the workspace.

5. **Error Display and Debugging:**

When a program is executed, any syntax or runtime errors are displayed in the terminal, helping developers to identify and fix issues quickly.

Use in This Project

In the **“Rock, Paper, Scissors Game using Python”** project, the **Command Prompt/Terminal** was used to:

- Run the Python file containing the game code.
- Display interactive messages between the user and the computer.
- View the program’s input and output in real-time.

Advantages

- Lightweight and fast for program execution.
- Eliminates the need for external applications.
- Provides direct control over the Python environment.

The **Command Prompt/Terminal** is an essential tool in Python programming that enables efficient code execution and debugging. Its simplicity, speed, and versatility make it an important part of every developer's workflow, especially for small and interactive projects like this one.

3.5 System Hardware and OS

Every software project requires a compatible combination of **hardware** and **operating system** to ensure smooth development, testing, and execution. The efficiency, stability, and compatibility of the system play a vital role in determining the overall performance of the software. For this project titled “**Rock, Paper, Scissors Game using Python,**” a standard computer system with moderate specifications was used to develop and run the program effectively.

The hardware configuration consisted of an **Intel Core i3 processor**, **4 GB RAM** (with 8 GB recommended for optimal performance), and **256 GB SSD or 500 GB HDD storage**. The display resolution used was **1366 × 768 pixels**, suitable for viewing and editing code comfortably. Standard **input devices** like a keyboard and mouse were used for code typing and execution. These specifications provided a stable and responsive environment for executing Python scripts without any lag or delay.

On the software side, the system was running on **Windows 10 (64-bit)** operating system, which provided compatibility with the latest versions of Python and related tools. The programming language used was **Python 3.11**, executed and edited using **Visual Studio Code (VS Code)** as the primary **Integrated Development Environment (IDE)**. The built-in **Command Prompt** was used to run the Python file and view the outputs in real time. Additionally, the project utilized Python's **random module**, which is included by default in the Python Standard Library, to generate random choices for the computer's move during gameplay.

In conclusion, the combination of suitable hardware and a reliable operating system created an efficient setup for the project. It ensured smooth execution, quick debugging, and a seamless development experience while working on the Python-based “Rock, Paper, Scissors” game.

3.6 Text Formatting & Documentation Tools

Text formatting and documentation tools play a vital role in creating well-organized, readable, and presentable project reports. These tools help in maintaining a professional layout, consistent structure, and clear documentation of the work done during the training period. Proper documentation ensures that the project details, coding explanations, and results are conveyed effectively to readers and evaluators.

In this project, **Microsoft Word** was used as the primary documentation tool for preparing and formatting the **Industrial Training Report**. It provided features such as font customization, text alignment, table creation, and page numbering that helped maintain a clean and structured layout throughout the report. The use of **styles and headings** made it easier to organize chapters and subtopics systematically, ensuring a professional appearance.

Additionally, tools like **Google Docs** and **PDF converters** were also utilized for editing, reviewing, and final submission of the report. These tools enabled real-time editing, easy sharing, and compatibility across different platforms. Screenshots and diagrams of the Python project were inserted and labeled properly using these documentation tools to make the content visually appealing and easy to understand.

Overall, the use of text formatting and documentation tools ensured that the report maintained a high standard of presentation. It allowed the project to be expressed clearly, both technically and visually, while highlighting the systematic effort and learning outcomes achieved during the training period.

CHAPTER 4

Project Description

4.1 Project Title

Rock, Paper, Scissors Game Using Python Programming”

This project is a simple yet engaging Python-based game that allows a user to play **Rock, Paper, Scissors** against the computer. It demonstrates the practical use of fundamental Python concepts such as **conditional statements, loops, functions, and the random module**.

4.2 Objective

The main objective of this project, **“Rock, Paper, Scissors Game Using Python Programming,”** is to design and develop a simple, interactive console-based game that allows users to play against the computer. The project aims to demonstrate the use of **basic Python concepts** such as input/output handling, conditional statements, loops, and random number generation in a practical application.

The key objectives of the project are as follows:

- To understand and implement **Python programming fundamentals** effectively.
- To apply **logical and decision-making concepts** for determining game outcomes.
- To develop a **user-friendly program** that interacts with players and provides instant results.
- To utilize the **random module** in Python for generating computer moves, ensuring fairness and unpredictability in gameplay.
- To enhance **problem-solving and coding skills** through the creation of a real-time interactive program.

In summary, the project’s primary goal is to strengthen the understanding of Python basics while developing a fun, functional, and logical game application that demonstrates programming efficiency and creativity.

The project also encourages analytical thinking and helps learners gain confidence in writing independent code.

4.3 Description of Project

The project titled “**Rock, Paper, Scissors Game Using Python Programming**” is a simple yet interactive console-based game designed to simulate the classic hand game between the user and the computer. The main purpose of this project is to demonstrate how core Python concepts can be applied to create an engaging and functional application.

The project uses Python’s **random module** to generate the computer’s choice from three available options — rock, paper, or scissors. The user is prompted to enter their choice through the keyboard using simple character inputs (r, p, or s). Once both inputs are received, the program compares them using **conditional statements** to determine whether the user wins, loses, or if the game ends in a draw.

The program structure is kept simple and easy to understand. It begins by importing the required module, defining user and computer choices, displaying both selections, and finally applying logic to display the result. The code execution takes place in a **command-line interface**, and the results are displayed instantly, making it an ideal beginner-level project for learning **decision-making and flow control** in Python.

This project demonstrates several important programming principles, such as user input handling, use of conditional and logical operators, and generating random outputs. It also improves the learner’s understanding of Python’s syntax, structure, and error handling.

Overall, the project successfully combines learning with fun. It shows how even a few lines of well-written Python code can be used to build an interactive and enjoyable program. The simplicity of the game makes it a perfect starting point for beginners to practice and strengthen their programming logic and creativity.

4.4 Tools and Technologies Used

Tools and Technologies Used

The project “**Rock, Paper, Scissors Game Using Python Programming**” was developed using a set of simple yet powerful tools and technologies that supported the coding, execution, testing, and documentation processes. These tools provided a smooth development environment and helped ensure that the program ran efficiently and accurately.

1. Programming Language – Python

Python was the primary programming language used in this project. It is a high-level, interpreted language known for its simplicity and readability. Python’s built-in functions, clear syntax, and extensive standard library made it an ideal choice for developing this

interactive game. The project specifically used **Python 3.11**, which provided better performance and modern language features.

2. Integrated Development Environment (IDE) – Visual Studio Code

Visual Studio Code (VS Code) was used for writing, editing, and testing the Python code. It provided features such as syntax highlighting, auto-completion, debugging tools, and an integrated terminal. These features enhanced the coding experience and allowed smooth execution of the program.

3. Command Prompt / Terminal

The **Command Prompt** (Windows) was used to run the Python program directly from the command line. It enabled the user to execute the .py file, view outputs, and debug errors efficiently. The terminal provided a fast and reliable environment for real-time program interaction.

4. Random Module

The **random module** is a built-in Python library used to generate random numbers or select random items from a list. In this project, it was used to make the computer's choice — rock, paper, or scissors — unpredictable, ensuring fairness and variety in gameplay outcomes.

5. Operating System – Windows 10

The project was developed and executed on **Windows 10 (64-bit)**. It provided a stable and compatible environment for Python, Visual Studio Code, and other required tools. The OS also supported smooth integration between the IDE and terminal.

6. Documentation Tools

Microsoft Word and **Google Docs** were used for preparing and formatting the Industrial Training Report. These tools helped in maintaining a professional structure, inserting screenshots, and ensuring the content was clear and well-presented.

4.5 Modules of the Project

The project “**Rock, Paper, Scissors Game Using Python Programming**” is a small-scale but structured application that consists of several logical modules. Each module performs a specific function, and together they form the complete workflow of the game. The modular design makes the program easier to understand, debug, and modify in the future.

1. Input Module

This module is responsible for taking the **user's input**. The player is prompted to choose one option among Rock, Paper, or Scissors by entering a corresponding character — r for Rock, p

for Paper, or s for Scissors. The input is stored for comparison with the computer's randomly generated choice. Proper validation ensures that the user enters a valid response.

2. Random Choice Module

This module uses Python's built-in **random module** to generate the computer's choice from the available options (Rock, Paper, or Scissors). The randomness ensures that each round of the game is fair, unpredictable, and not biased toward the player or the computer.

3. Comparison and Decision Module

The core logic of the program is handled in this module. It compares the user's choice with the computer's choice using **conditional statements**. Based on the comparison, the program decides whether the result is a **Win, Lose, or Draw**. This module demonstrates the use of if-else conditions and logical operators effectively.

4. Output Module

This module is responsible for displaying the results to the user. It prints both the user's and computer's selections in a readable format and announces the outcome of the round. The output module ensures clear, well-formatted, and user-friendly messages.

5. Main Control Module

The main control module integrates all the above modules and ensures smooth program flow. It initializes the random module, calls each function in order, and controls the execution from start to finish. This module acts as the **driver** of the program, managing the overall logic and sequence of events.

4.6 Algorithm/ Working Steps

The **Rock, Paper, Scissors Game using Python Programming** operates through a clear sequence of steps that define the program's logic and flow.

1. **Start the Program:** Import the random module to enable random selection for the computer's move.
2. **Display Instructions:** Show the user how to play and which keys correspond to Rock, Paper, and Scissors.
3. **Take User Input:** Prompt the user to enter their choice (r, p, or s) and store it in a variable.
4. **Generate Computer Choice:** Use random.choice() to randomly pick Rock, Paper, or Scissors for the computer.
5. **Display Choices:** Print both the user's and the computer's selected options.

6. **Compare the Choices:** Use conditional statements (if-elif-else) to determine the result based on standard rules (Rock beats Scissors, Scissors beats Paper, Paper beats Rock).
7. **Display the Result:** Print whether the user wins, loses, or if the game is a draw.
8. **End the Program:** Display a closing message and stop the program or prompt for a replay.

4.7 Sample Output

Below is a sample output of the **Rock, Paper, Scissors Game using Python Programming**. This demonstrates how the program interacts with the user and displays results based on both user and computer choices.

Example 1: When User Wins

```
Enter your choice (r for Rock, p for Paper, s for Scissors): p
You chose Paper
Computer chose Rock
You Win!
```

Example 2: When Computer Wins

```
Enter your choice (r for Rock, p for Paper, s for Scissors): s
You chose Scissors
Computer chose Rock
You Lose!
```

Example 3: When It's a Draw

```
Enter your choice (r for Rock, p for Paper, s for Scissors): r
You chose Rock
Computer chose Rock
It's a Draw!
```

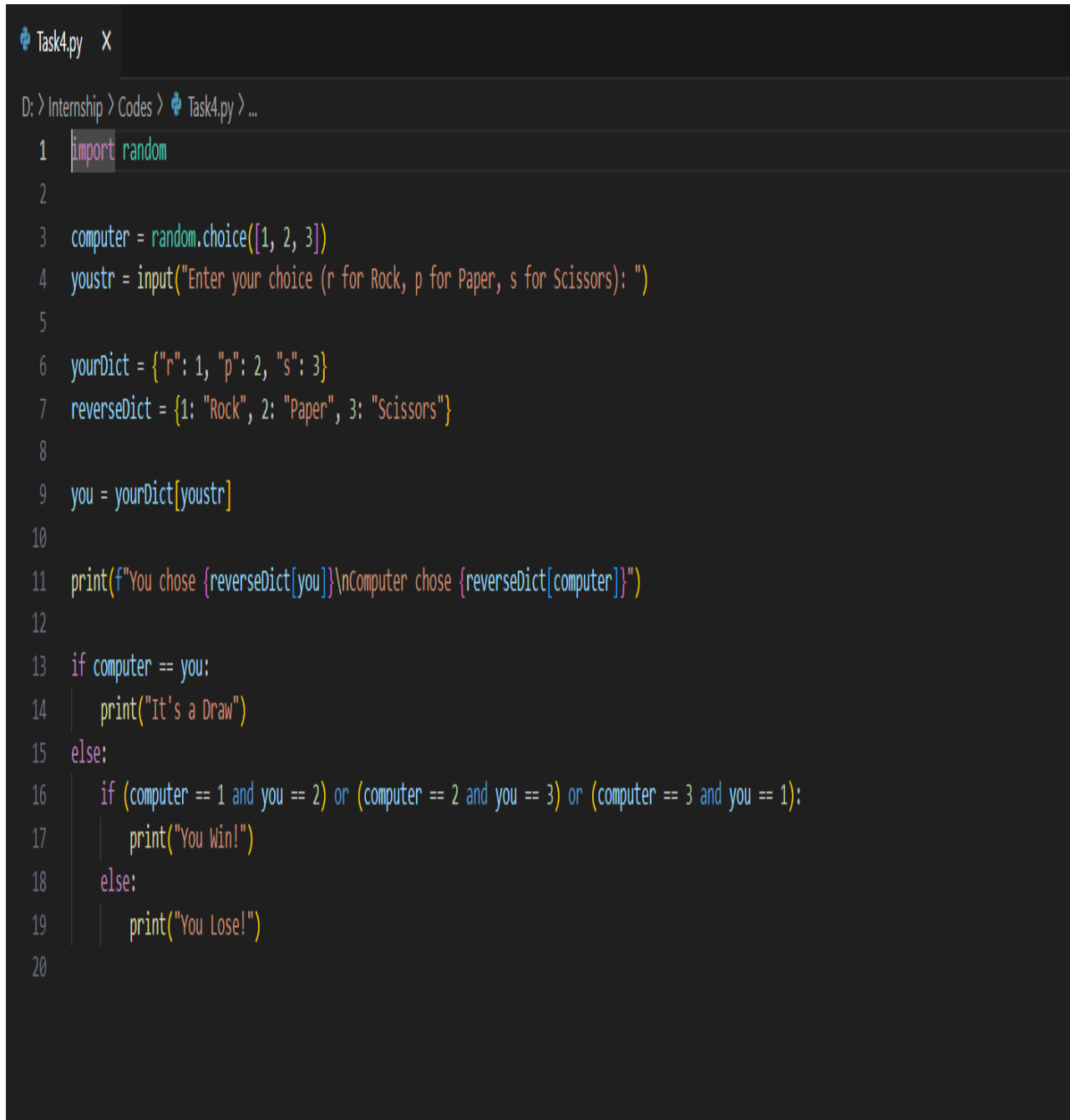
4.8 Learning Outcomes

During the development of the **Rock, Paper, Scissors Game using Python Programming**, several important technical and conceptual skills were learned. These outcomes reflect the understanding and practical knowledge gained throughout the project.

1. **Improved Understanding of Python Syntax:** Learned the correct use of Python statements, indentation, variables, data types, and input/output functions.
2. **Practical Knowledge of Conditional Logic:** Gained experience in applying if-elif-else conditions to control program flow and make logical decisions.
3. **Use of Built-in Python Modules:** Understood how to import and use Python's random module for generating unpredictable outcomes.
4. **Hands-on with User Interaction:** Learned to design simple yet effective console-based interfaces for user input and output display.
5. **Debugging and Problem-Solving:** Enhanced the ability to identify logical errors, test different scenarios, and improve program reliability.
6. **Algorithm Design and Flow:** Acquired the skill to break down a problem into clear working steps and implement them in code.
7. **Documentation and Presentation:** Learned how to write clean, readable code and document the project in a structured and professional format.

CHAPTER 5

SCREENSHOTS



```
Task4.py X
D: > Internship > Codes > Task4.py > ...
1 import random
2
3 computer = random.choice([1, 2, 3])
4 youstr = input("Enter your choice (r for Rock, p for Paper, s for Scissors): ")
5
6 yourDict = {"r": 1, "p": 2, "s": 3}
7 reverseDict = {1: "Rock", 2: "Paper", 3: "Scissors"}
8
9 you = yourDict[youstr]
10
11 print(f"You chose {reverseDict[you]}\nComputer chose {reverseDict[computer]}")
12
13 if computer == you:
14     print("It's a Draw")
15 else:
16     if (computer == 1 and you == 2) or (computer == 2 and you == 3) or (computer == 3 and you == 1):
17         print("You Win!")
18     else:
19         print("You Lose!")
20
```

OUTPUT

```
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe d:/Internship/Codes/Task4.py
Enter your choice (r for Rock, p for Paper, s for Scissors): r
You chose Rock
Computer chose Paper
You Lose!
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe d:/Internship/Codes/Task4.py
Enter your choice (r for Rock, p for Paper, s for Scissors): p
You chose Paper
Computer chose Paper
It's a Draw
PS C:\Users\hp> & C:/Users/hp/AppData/Local/Programs/Python/Python313/python.exe d:/Internship/Codes/Task4.py
Enter your choice (r for Rock, p for Paper, s for Scissors): s
You chose Scissors
Computer chose Rock
You Lose!
PS C:\Users\hp> █
```

CONCLUSION

The completion of the **Rock, Paper, Scissors Game using Python Programming** marks a significant milestone in understanding and applying the fundamental concepts of programming, logic building, and algorithm design. Throughout this project, I have gained both theoretical insight and practical experience in developing a simple yet interactive Python application. The entire process—from planning the structure of the game to implementing it and finally analysing its performance—has been a valuable learning journey that enhanced my problem-solving and technical skills.

The project primarily focused on developing a small-scale interactive game that demonstrates the use of Python programming in a real-world scenario. Although the game itself appears simple, it incorporates several core programming principles such as conditional statements, loops, random number generation, user input handling, and program output formatting. The process of developing this game provided a deeper understanding of how logical decisions are implemented in software and how user interaction can be managed effectively in a command-line environment.

One of the most important aspects learned during this project is the **use of modular programming** and the significance of breaking down complex problems into smaller, manageable parts. The implementation of the random module to simulate the computer's moves introduced the concept of unpredictability in programming, which is a crucial component of game logic and many real-world systems. This gave me exposure to how built-in Python libraries simplify programming tasks and allow developers to achieve complex behaviour with minimal code.

Additionally, the project enhanced my understanding of **algorithmic thinking**. Designing the logic behind the game required careful analysis of all possible scenarios—when the user wins, loses, or ties with the computer—and translating them into conditional statements. This approach taught me how to design efficient algorithms that handle every possible case without unnecessary redundancy. It also reinforced the concept of logical operators, control structures, and flow of execution in Python.

The use of **Visual Studio Code (VS Code)** as the development environment provided an opportunity to learn how Integrated Development Environments (IDEs) simplify the coding process through features like syntax highlighting, debugging support, and code execution tools. I also learned how to run Python scripts through the **Command Prompt or Terminal**, manage file directories, and handle execution errors effectively. These experiences contributed to developing a more professional approach to software development and deployment.

Working on this project also improved my understanding of **software documentation** and the importance of presenting a project systematically. Writing structured sections such as the introduction, objective, algorithm, tools used, and learning outcomes helped me recognize the value of clear documentation in software projects. Proper documentation ensures that anyone reading or maintaining the project can easily understand its purpose, functionality, and structure.

Another important outcome of this project was the improvement of my **debugging and testing skills**. During development, various logical and syntax errors were encountered. Debugging these issues required patience, analysis, and the ability to trace the flow of the program. This not only improved my logical reasoning but also taught me how to test individual parts of a program to ensure correctness and efficiency.

From a broader perspective, this project helped me appreciate how even a small-scale program like Rock, Paper, Scissors demonstrates the essence of software design and user interaction. It mirrors the decision-making process seen in more advanced AI-driven applications and highlights how randomness, probability, and condition-based logic play a crucial role in digital systems.

Furthermore, this project strengthened my confidence in Python as a versatile and beginner-friendly programming language. I realized how easily Python allows developers to express ideas with minimal syntax while maintaining high readability. This makes Python not only ideal for small projects but also powerful enough for large-scale applications in data science, web development, and artificial intelligence.

Overall, this project served as a strong foundation for developing more advanced software solutions in the future. It provided a stepping stone toward understanding real-world programming environments, basic game development principles, and algorithmic thinking. The

experience gained will be extremely valuable for future academic projects, internships, and professional work in the IT and software development fields.

In conclusion, the **Rock, Paper, Scissors Game Project** successfully met its objectives by implementing a working model of an interactive Python-based game that demonstrates logical reasoning, user interaction, and computational thinking. It not only enhanced my technical proficiency in Python but also nurtured creativity, analytical thinking, and attention to detail. This project reaffirmed my interest in software development and strengthened my motivation to explore advanced areas like artificial intelligence, data structures, and machine learning in the future.

Through this experience, I have understood that programming is not just about writing code—it is about solving problems, designing logical systems, and creating efficient solutions that can interact intelligently with users. The lessons learned during this project will continue to guide my academic and professional journey as I aim to become a proficient software developer and contribute to the evolving field of technology.

FUTURE SCOPE

The **Rock, Paper, Scissors Game using Python Programming** project has significant potential for further improvement and expansion. Although the current version fulfills its basic purpose, there are numerous possibilities to enhance its features, performance, and user experience. The following points describe the various directions in which this project can be developed in the future.

Graphical User Interface (GUI) Integration: The present game operates through a command-line interface, which limits its visual appeal. In future versions, a Graphical User Interface (GUI) can be developed using Python libraries such as Tkinter, PyQt, or Pygame. This will allow users to interact with the game through buttons, icons, and animations, making it more engaging and visually attractive. A GUI will transform the simple text-based game into a professional, interactive desktop application.

Multiplayer Mode: At present, the game allows a single player to play against the computer. Introducing a multiplayer mode will make the game more interesting and competitive. Two players could play using the same system or through online connectivity using socket programming or web APIs. Online multiplayer functionality will allow users to challenge each other remotely, enhancing both the entertainment value and the developer's understanding of networking and client-server architecture.

Score Tracking and Game History: Adding a scoring system will make the game more interactive. The program can record the number of wins, losses, and draws for each player. This data can be stored in a text file, CSV file, or a database to maintain a record of player performance. Leaderboards can also be introduced for multiplayer versions, making the game more competitive. This addition will also give practical knowledge of file handling and data storage.

Artificial Intelligence (AI) Implementation: Currently, the computer's moves are generated randomly. In future updates, AI can be used to analyze player behavior and predict moves. Machine learning models can be trained to adapt based on player choices, making the computer a

smarter opponent. Libraries like Scikit-learn, TensorFlow, or PyTorch can be utilized to integrate AI capabilities, thereby enhancing both gameplay and technical learning.

Voice Command Integration: Future versions of the game can include voice recognition features using libraries such as SpeechRecognition and PyAudio. Instead of typing, players could simply speak their choices, making the game more natural and interactive. This feature would also improve accessibility for users with physical disabilities and introduce developers to speech processing concepts.

Cross-Platform and Mobile Compatibility: The current version runs on desktops only. Expanding it to web or mobile platforms using frameworks like Flask, Django, or Kivy will make the game accessible from anywhere. This would increase its usability and reach while giving practical exposure to web and mobile application development.

Enhanced User Experience: The game can be made more appealing by adding sound effects, animations, and background music. Features such as theme selection, difficulty levels, and customization options can be provided to improve user satisfaction. A start menu, settings option, and result summary screen can make the interface look professional and polished.

Data Analytics and Visualization: In the future, data visualization libraries such as Matplotlib or Plotly can be used to show player statistics and performance charts. Players could view their winning trends, favorite choices, and improvement over time. This not only makes the game informative but also introduces practical experience in data analysis and visualization.

Cloud Integration: Integrating cloud storage can enable players to create profiles and save their scores online using Firebase, AWS, or Google Cloud. This will allow users to log in from any device and continue their game. Such an enhancement provides knowledge of real-world technologies like cloud databases and authentication systems.

Educational Use: This project can be converted into an educational tool for beginners learning Python. By explaining the logic, functions, and flow of the game, learners can understand programming concepts easily. It can be used in coding workshops or training sessions to demonstrate how simple logic can be used to build interactive applications.

Advanced Game Mechanics: The game can include advanced features such as different levels (Easy, Medium, Hard), time limits, or bonus points. Adding such elements will make the gameplay more exciting and challenging, while giving the developer exposure to event handling and dynamic logic implementation.

Security and Error Handling Enhancements: As the game grows to include online features, implementing proper error handling and security will become important. Techniques such as input validation, data encryption, and exception handling can be introduced to make the application more secure and reliable.

Summary: The future scope of the Rock, Paper, Scissors Game is vast and promising. By adding GUI, AI, cloud support, multiplayer capability, and analytics, this simple Python program can evolve into a smart, modern, and interactive gaming application. These enhancements will not only improve the user experience but also provide the developer with exposure to real-world programming concepts such as artificial intelligence, data visualization, and software design. Thus, this project forms a strong foundation for continuous learning, innovation, and technological advancement.

REFERENCES

1. Mark Lutz, *Learning Python*, 5th Edition, O'Reilly Media, 2013.
2. Al Sweigart, *Automate the Boring Stuff with Python*, No Starch Press, 2019.
3. John V. Guttag, *Introduction to Computation and Programming Using Python*, MIT Press, 2016.
4. Python Official Documentation, *Python 3.11*, <https://docs.python.org/3/>
5. W3Schools Python Tutorial, <https://www.w3schools.com/python/>
6. GeeksforGeeks Python Programming Tutorials,
<https://www.geeksforgeeks.org/python-programming-language/>
7. TutorialsPoint Python Tutorial, <https://www.tutorialspoint.com/python/index.htm>
8. Python Module Reference – Random,
<https://docs.python.org/3/library/random.html>
9. Visual Studio Code Official Documentation, <https://code.visualstudio.com/docs>
10. Stack Overflow Discussions on Python Projects,
<https://stackoverflow.com/questions/tagged/python>